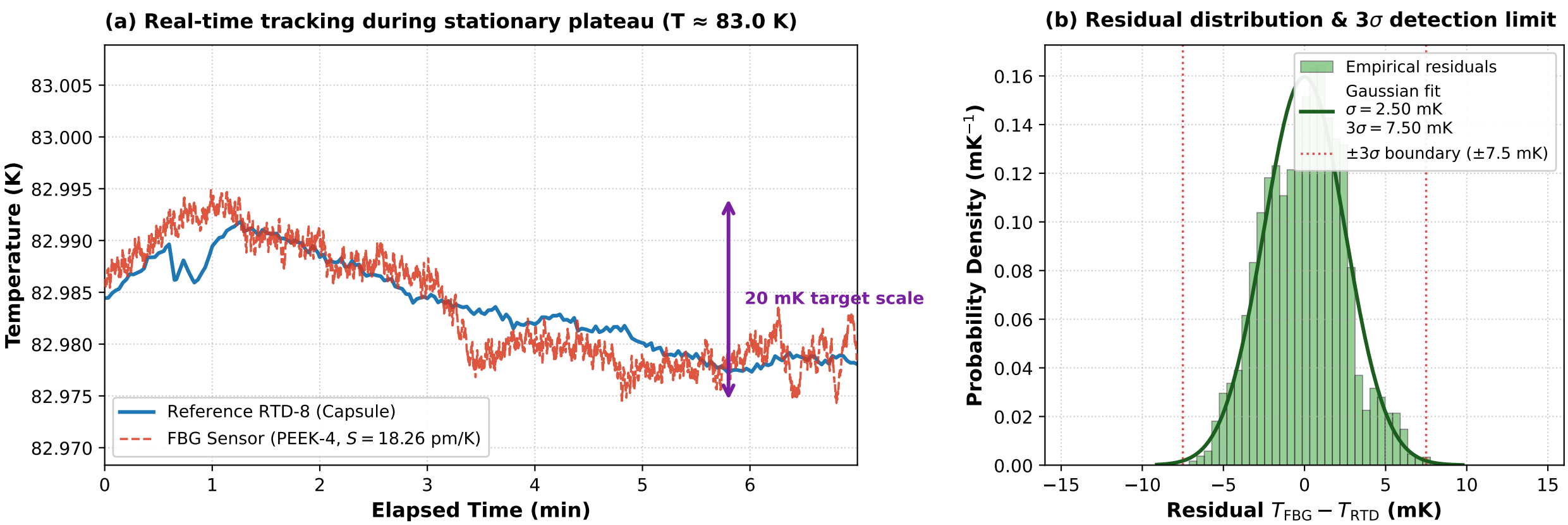

(a) Real-time tracking during stationary plateau (T ≈ 83.0 K)
Temperature (K)
83.005
83.000
82.995
82.990
82.985
82.980
82.975
82.970
0
1
2
3
4
5
6
Elapsed Time (min)
20 mK target scale
Reference RTD-8 (Capsule)
FBG Sensor (PEEK-4, S = 18.26 pm/K)
(b) Residual distribution & 3σ detection limit
Probability Density (mK⁻¹)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−15
−10
−5
0
5
10
15
Residual T_FBG − T_RTD (mK)
Empirical residuals
Gaussian fit
σ = 2.50 mK
3σ = 7.50 mK
±3σ boundary (±7.5 mK)